

RAMAIAH INSTITUTE OF TECHNOLOGY, BANGALORE-560054

DEPARTMENT OF CHEMISTRY

Sub: Engineering Chemistry, Code: CYE12, Course: I Sem B E (EEE stream)
 CIE Test: 2, Term : 30.11.2022 to 21.03.2023 Max. Marks: 30 Time : 60 min, Credit: 3:0:0
Instructions: Answer any two full questions. Each question carries 15 marks

Q.No.		Marks	CO's
1	a) Define glass transition temperature? Explain any two factors affecting T _g of polymers?	5	CO3
	b) Define the term liquid crystal. Discuss Classification of liquid crystals with suitable examples.	5	CO3
	c) Explain the synthesis of nanomaterials by solution combustion method with suitable example.	5	CO4
2	a) Write the synthesis, properties and applications of Teflon.	5	CO3
	b) What is a homologous series? Describe the liquid crystalline behaviour in homologous series compound by taking an example of PAA.	5	CO3
	c) State Beer-Lambert's law. Explain the instrumentation of colorimeter.	5	CO4
3	a) Give reasons: i) Teflon coated utensils are used for cooking or baking. ii) All simple organic polymers are not monomers. iii) Plexi glass is used in the manufacture of lenses.	5	CO3
	b) Describe the determination of the pH of beverages using Glass electrode.	5	CO4
	c) What are nanomaterials and give examples. Outline any four applications of nanomaterials.	5	CO4